

The PAMalyzer (v 0.0.1)

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WORK IN PROGRESS!

This is the user manual for ▲ version xxx of the PAMalyzer program. The PAMalyzer is based on AviaNZ, a software for bioacoustic analysis (<http://www.avianz.net>). There are so many changes compared to AviaNZ that I decided to name it differently. The most obvious are (i) the ability to classify your recordings with the BirdNET classifiers, and (ii) several features to ease the analysis of the BirdNET results with filtering and sorting options. The purpose of this manual is to help you get started with using the PAMalyzer. Furthermore a 'Cheat Sheet' is provided which you can find in the same directory as this manual or within the program under Help.

Quick-Start

1. **Set operator:** On first startup PAMalyzer will ask you to enter an operator name for future reference. More information in [Getting Started](#).
2. **Open an audio file:** If you did not use PAMalyzer before an example file will be loaded. You can use the *Open sound file* option in the *File* menu or press **Ctrl+O**. See [Open Audio Files](#) for details.
3. **Choose a database:** All annotations in PAMalyzer are stored in a database. The default database will be good for first tests. Choose another database via *Annotations>Database>Choose database* if you start productive work (see the [Database](#) section for more information).
4. **Classify recordings:** To classify your audio files launch BirdNET via *Analysis>Classify recordings with BirdNET*. Find more detailed information in the [Classify Recordings with BirdNET](#) section.
5. **Review detections:** PAMalyzer offers plenty of options for purposefull reviewing and filtering of detections. See the [Review Classifier Detections](#) section for a full workflow.

6. **Export results:** Export annotations and summary Excel files via the *Export* options in the *Annotations* menu. Refer to the `namerefsec:outputs` section for the export formats.

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1 Introduction

The PAMalyzer enables you to:

- Show and navigate through spectrograms of audio files
- Manually label bird calls in audio files
- Use the BirdNET classification models on your data
- Review the results of the BirdNET classification process

We will walk you through them individually. If you are using PAMalyzer the first time the section [Getting Started](#) is a good starting point.

2 Installation and Setup

For Windows users there is a downloadable installation file available on the github page <https://github.com/FloMee/PAMalyzer/releases>. This should work for most users.

If you plan on using the software under Linux or MacOS, follow the instructions in the README to install the software from source.

On Windows some files will be placed in your home directory, which may be somewhere like `C:/Users/username/AppData/Roaming/PAMalyzer`. On Mac and Linux they will be in `~/.PAMalyzer`.

3 Getting Started

On the first startup the program will ask you to give a name for the operator. This is useful to keep track of who has looked at different files. After confirming the main interface (figure 1) shows up. By default a demo audio file is shown. If you want to move to a new directory, either use the ‘..’ option to navigate around your computer’s file system, or use the *Open sound file* in the *File menu*.

***Note:** PAMalyzer can load audio files in wav format. Other audio files should be converted to wav format. PAMalyzer assumes that audio files are recordings in mono (single microphone) sound. If there are other channels, only the first one is loaded. Very long recordings can be used, but it is generally easier if they are split into shorter files.*

3.1 The Main Interface

The name of the current file is shown in the title of the window (1). Below that there are four separate areas of the screen. Each has its name in a blue bar. They are:

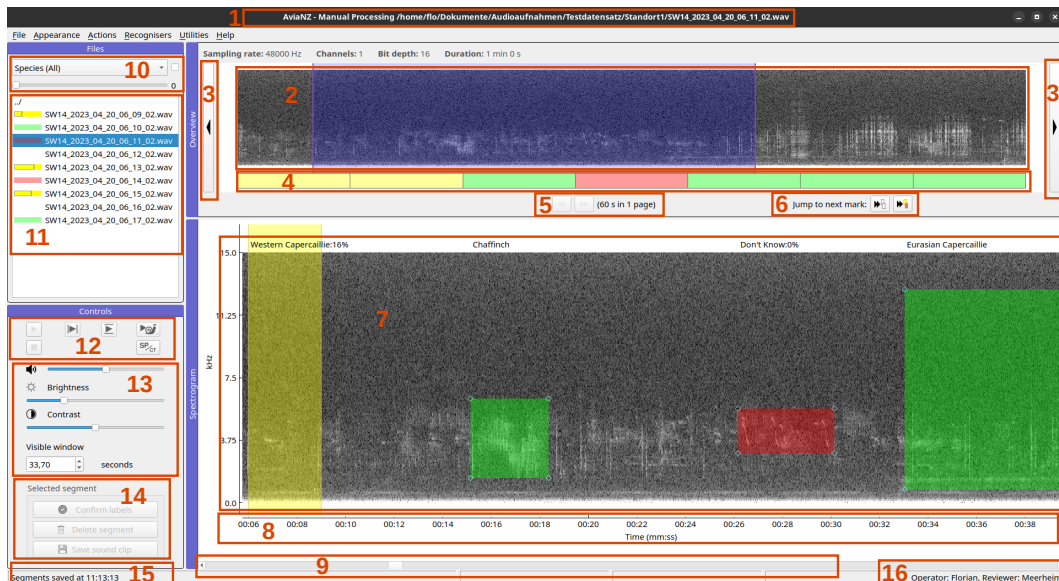


Figure 1: The interface of the *Manual Processing* mode with its different components.

Overview (2–6) This area shows a spectrogram of the first 5 minutes of the current audio file (labelled 2 in the figure). The blue section indicates the part you are looking at in the Spectrogram plot below. It can be moved by using **Left** and **Right** arrow keys, dragging the centre of the box, by clicking on the left and right arrow buttons (◀, ▶) next to the spectrogram (3) or by using the buttons in the bar below the spectrogram (4). It can be resized by dragging the ends of the box. Below the bar with buttons there are double arrow buttons (◀◀, ▶▶, labelled 5 and 6). With the buttons labelled 5 you can open the next 5 minutes page of the audio file, provided it exists. The other buttons (6) you can use to jump with the blue section to the next annotation segment.

Spectrogram (7–9) In this area the spectrogram of the blue section in the Overview area is shown. Here you can add and modify annotations. The axes for the image shows frequencies on the vertical axis and time on the horizontal axis (8). The time is true time if PAMalyzer can read that from the file, and starts from time 0:00 otherwise. There is also a scroll bar to move through the current page (9). The way in which you annotate recordings is described in the section [Annotate Recordings](#).

Files (10–11) This is the list of audio files (11) in your current directory. Use the **Up** or **Down** arrow buttons on your keyboard to navigate through the file list. You can also double-click on a file to select and open it. The file icons next to the file names indicate the status of the audio file. This is described in detail in the section [Colour Codes](#). There are several filtering options (10) to reduce the file list. These are described in [Filtering Audio Files](#).

Controls (12–14) The buttons in this area are referring to the Spectrogram area. You

can playback the audio file (12), modify the appearance of the plots (13) and process any segment that is currently selected (14). For more details, see [Controls](#).

There is an optional fifth area, which is the Amplitude plot. If you want to see it, select *Show amplitude plot* in the *View* menu. Select the option again to hide it. You can also hide the *Files* and *Overview* area in the same way. PAMalyzer will remember your choice in future uses.

***Note:** You can drag the five screen areas around and reorder them if you wish, by dragging the blue bar on the top or left of them with the name of the section in it. You can also make them into their own windows by double-clicking on the blue bar. If you decide that you made a mistake doing that, then there is an option in the Actions menu to Put docks back that returns them to the original configuration.*

There are two other parts to the interface:

Menu The menu at the top of the screen.

Status Bar At the very bottom of the screen there is an area that gives any status updates from the program (15), and on the right, information about the currently selected Reviewer and Operator (16).

3.2 Cheat Sheet

You can find a Cheat Sheet with the most important navigation options, especially shortcuts in the *Help* menu under *Cheat Sheet*. Especially in the review process after the classification of your audio files with BirdNET it can help a lot.

4 Open Audio Files

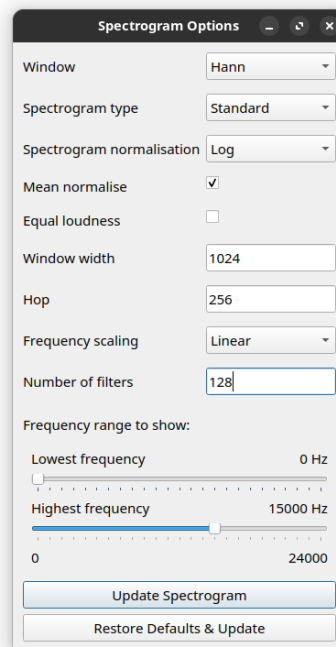
4.1 Spectrogram

- The main spectrogram plot (7) shows a section of the current audio file. The part you are looking at is highlighted in blue in the overview spectrogram (2).
- The axes of the spectrogram are time on the horizontal axis (8) and frequencies on the vertical axis.
- The times will be true times if file names contain the time stamp in the format `<year><month><day>_<hour><minute><second>`, or time from start of file otherwise.
- Sometimes spectrograms don't look good initially, for example because of high noise. You can modify the brightness and contrast of the spectrogram using the two sliders in the Controls section (13).

- You can also use a different colour scheme, and invert the colour map (swap black and white), by choosing the relevant options in the *View* menu.
- To change the computation of the spectrogram you can use the *Change spectrogram parameters* from the *View* menu. The options are described in detail in section 4.1.1.
- It is possible to display information about where the mouse is pointing on the spectrogram (time, frequency, energy value) below the axis by choosing *Show pointer details in spectrogram* from the *View* menu, and to switch it off the same way.

4.1.1 Changing the Spectrogram Computation

- Selecting *Change spectrogram parameters* from the *View* menu will produce the following dialog box:



- You can change some of the parameters that are used to produce the spectrogram:
 - The window function (default: Hann) controls how the spectrogram combines sounds across the time range.
 - Mean normalisation and equal loudness try to make the spectrogram energies more even.

- Multitapering uses multiple windows to make a better estimate of the spectrogram, but performs a lot of computation, and can therefore be slow.
- The window width and hop size control how much of the audio file makes up one spectrogram bin, and how they overlap. A big window size will improve frequency resolution, but reduce time resolution, and vice versa. The hop size controls how much the spectrogram time bins overlap. It is computationally more efficient if these numbers are powers of 2 (such as 256, 512, 1024).
- The frequency range sliders let you change the visible frequency range in the main spectrogram window. This can be useful if you want to focus on only part of the range.

4.1.2 Denoising

- When audio files are particularly noisy, it can be helpful to remove some of that noise. PAMalyzer currently provides three ways to do this (via the *Denoise* option in the *Actions* menu):

Wavelets This is our main method, and tries to preserve the bird call perfectly.

Bandpass Suppresses all frequencies outside a restricted range (which you specify) so that you can concentrate on the frequency range where calls that you are interested in can be seen.

Butterworth bandpass Another way to compute a restricted frequency range.

- You can save the denoised audio files, and undo the denoising if it does not help.

5 Navigation

Navigate through Files To quickly navigate through the list of files and the files themselves it is highly recommended to use the arrow keys on the keyboard: Use the **Up** and **Down** arrow keys to open the previous or next audio file in the file list. Use the **Left** and **Right** arrow keys to move backward and forward through the audio file.

But there are a lot of more options:

- To open an audio file:
 - Use the **Up** or **Down** arrow keys to open the next or the previous file in the file list.
 - Use *Open audio file* in the *File* menu if you want to open an audio file in another directory.
 - Double Click on a file in the list of files.
 - Double Click on a folder will open that folder; the ‘..’ option at the top of the list takes you up one directory.
- To move within a file:

Note: If some of the filter options in the Files area (10) are applied, some of the files might be hidden within the file list. In this case the **Up** and **Down** arrow keys apply only for the visible files. The shortcut **Alt + Up/Down** in contrast includes also the hidden files, so that you can still reach them.

- Press the **Left** or **Right** arrow keys.
- Drag the blue highlight in the overview spectrogram.
- Click on any of the boxes in the bar below the overview spectrogram.
- Click on the left or right arrows on the left and right of the overview spectrogram.
- Drag the scroll bar below the spectrogram.

Note: According to the set page size (defaults to 5 minutes) in the [Interface settings](#) long files are split into several pages. To open the next or previous page of an audio file use the double arrow buttons (5) below the overview spectrogram or the shortcut **Shift + Left/Right**. More details in: [Moving Through Long Files](#).

- To change the amount of the file that you can see (the visible width):
 - Drag the ends of the blue highlighted section in the overview spectrogram.
 - Change the *Visible window width (seconds)* in the Controls dock by clicking on the up and down arrows, or typing in a new number.
- You can also view a restricted amount of the spectrogram by reducing the visible frequency band, see Section [Changing the Spectrogram Computation](#).

Moving Through Long Files According to the set page size (defaults to 5 minutes) in the [Interface settings](#) long files are split into several pages.

- To move to the next or previous page:
 - Use the double arrow buttons labelled (5) or
 - press **Shift + Left/Right** arrow keys.
- The program tells you which page you are currently on, and how many there are in total.
- You can change the page size and the amount of page overlap by choosing the relevant options in the [Interface settings](#) in the *View* menu.
- The times on the axis below the spectrogram show locations in the full file.
- Note that operations like denoising and segmentation (described in the section [Analysis Menu](#)) apply to the visible 5 minute portion of the file, not the whole file.

6 Annotate Recordings

6.1 Overview

The aim of the labelling process is to add further information to the audio file. For this purpose PAMalyzer uses labeled segments. A segment is a specific range in time and frequency within the spectrogram. Together with the information about time and frequency a species label (e.g. a bird species) and a call type label (for specific call types of the species) can be stored.

- To create segments, click and drag with the left mouse button on the Spectrogram. (The action of left and right mouse buttons can be swapped over in the *Interface settings*.)
- To select segments, click on them with the right mouse button (pressing control on the keyboard when clicking on a Mac). The segment will turn blue when it is selected.
- If you find that the colours make it hard to see the data underneath the boxes you can make them transparent using the *Make dragged boxes transparent* option under *Annotation* in the *Interface settings*. You can also choose different colours there if you have particular preferences.
- Segments are saved automatically, so that you can't lose your work.

6.2 Create and Label Segments

- There are three ways to create a segment. You change which of them to use in the *Mouse settings* of the *Interface settings* in the *View* menu:
 - (*default*) Drag a limited frequency band box (i.e., click and hold the mouse button and drag the mouse to the correct end point in both time and frequency).
 - Start and stop a limited frequency band by clicking (i.e., click once at the correct time and frequency for the start of a segment, and then again at the time and frequency for the end).
 - Start and stop a full frequency band by clicking (i.e., click once at the correct time for the start of a segment, move the mouse to the end, click again, the box covers all frequencies).
- When you create a new segment, a drop-down menu will appear asking you to choose a label for that segment.
 - To choose the type of bird, click on the name.
 - If the species isn't in the list, move to 'Other'.

- If it isn't in the second list, at the bottom of that list is a selectable box (it probably says 'Albatross'). Clicking on it provides the complete list of birds.
- If there is something missing from there, choose 'Other' from it, which is at the very end. It will ask you to enter a name, as Genus (Species); e.g., 'Kiwi (Little Spotted)'.
- If there is only a single example of the genus, you can miss out the species, e.g., 'Kakapo'.
- The new name you add will then appear in the list.
- If you click anywhere on the screen except on a bird name in the menu then the menu will disappear and the bird type will be labelled as 'Don't Know'.
- The bird list that we are using is based on the one that DOC use, and is meant to cover all the bird species that we know about in New Zealand.
- You can add new species individually using 'Other' in the menus.

Note: *The original version of PAMalyzer comes with a bird list for New Zealand. In the list the species names are given as 'Genus (Species)'; e.g. 'Kiwi (Little Spotted)'. Given it in that way a submenu for every genus is created with the species in it. Although this works great for common species names in English, it doesn't in other languages. It is possible to use other bird lists (that do not necessarily have to follow this formatting) using the [Interface settings](#).*

- By default the lists of bird names updates dynamically, so that bird types you have chosen appear at the top of the list. If you don't like that, then you can disable it in the [Interface settings](#).
- When creating a segment, you can give it the same label as the previous segment by pressing the **Shift** key on the keyboard when you create the segment. The program will then give the segment the same label as the previous box that you labelled, without showing you the list. This is very useful when there is one bird calling repeatedly.
- You can also show that you are uncertain by pressing the **Ctrl** button when you click to segment (command button on a Mac computer). The names of the birds will then have a question mark after them.
- You can choose whether or not to allow more than one species per segment. This can be useful for recordings of the dawn chorus and where you just want to label presence of a set of species. Use the *Default to multiple species* in the [Interface settings](#) to allow this. Then, when you choose a bird from the list it will be ticked, and the menu will not close automatically, allowing you to make multiple selections. To unselect something, click on it again. When you make a new segment, 'Don't Know' is selected by default. Choosing any other option deselects 'Don't Know'. You can reselect it if you do want that label as well.

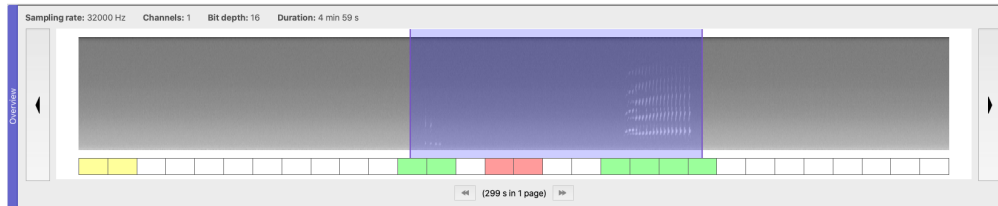
- PAMalyzer also allows you to specify a call type. This could be the difference between male and female vocalisations for e.g., kiwi, or different calls types such as ‘more-pork’ or ‘cree’ for ruru. You can add specific call types manually. To swap between seeing the species and seeing the call type label use the SP/CT button () on the right of area 12 of the Controls panel or press the **Tab** key.

6.3 Update Segments

- If you select a segment that already exists by clicking on it then it will turn blue. Click on it again and the menu will reappear so that you can correct mistakes. This will be either the species label or the call type label depending on the SP/CT button.
- Segments have blue diamonds at the corners, so that you can resize them, or you can move the whole segment by dragging it (this works even when it isn’t selected).
- Segments can be deleted by selecting them (so that they turn blue) and then clicking on the *Delete Current Segment* in the Controls or pressing the **Delete** or **Backspace** key on the keyboard.
- To delete all of the segments from the current file, use the *Delete all segments* option in the *Actions* menu or the shortcut **Ctrl + D**.
- You can disable the making and updating of segments (to avoid making segments by mistake) by selecting *Make read only* from the *View* menu.

6.4 Colour Codes

- The segments that are drawn on the screen have different colours. These colours are changeable in the *Interface settings*, but by default are:
 - Blue** This segment is currently selected. You can play it by pressing the buttons in the Controls area (see Section 12.1.4), or give it a new label by clicking on it again, or delete it.
 - Green** This segment has been labelled with a bird name and a confidence value of 100. These are either manually created segments or the manually confirmed results of an automatic filter.
 - Yellow** This segment has been labelled with a bird name and a confidence value between 0 and 100. These are either manually created segments with a question mark (by pressing **Ctrl** while dragging the box), or the row result of an automatic filter.
 - Red** This segment has been labelled as ‘Don’t Know’. Its confidence value is 0.
- These colours match the rectangles underneath the Overview:



- For each 10 second segment of the file, these boxes are:
White if there are no segments, or
Red if there are 'Don't know' segments, or
Yellow if there are questionable segments, or
Green if all the segments in that section are labelled with definite species.
- You can click on those boxes, and they will update the main spectrogram plot to show that section of the file. This is a useful way to move through the file quickly.
- The described color scheme also matches the colors of the file icon in the file list.
White if there are no segments in the file or directory, or
Red if there are 'Don't know' segments, or
Yellow if there are questionable segments, or
Green if all the segments in the file or directory are labelled with definite species.
- Moreover, the grey box in the file icon indicates the maximum confidence value found in the segments of the file.

***Note:** The filtering options for the file list (10) also effect the file icon. If the segments in the file do not match the set conditions the file icons turn white. You can use this feature to quickly search for candidate recording after the classification process with BirdNET. The workflow with BirdNET is described in the section [Classify Recordings with BirdNET](#)*

7 Classify Recordings with BirdNET

An overview of the workflow with BirdNET is shown on figure 2. It is divided into several steps, beginning with the classification process of the audio files, continuing with the filtering and reviewing of the detections coming to the summary of the results.

To start the analysis of your audio files with BirdNET you first have to specify different parameters. You can reach the input form by clicking on *Recognisers* → *Classify recordings with BirdNET* in the menu.

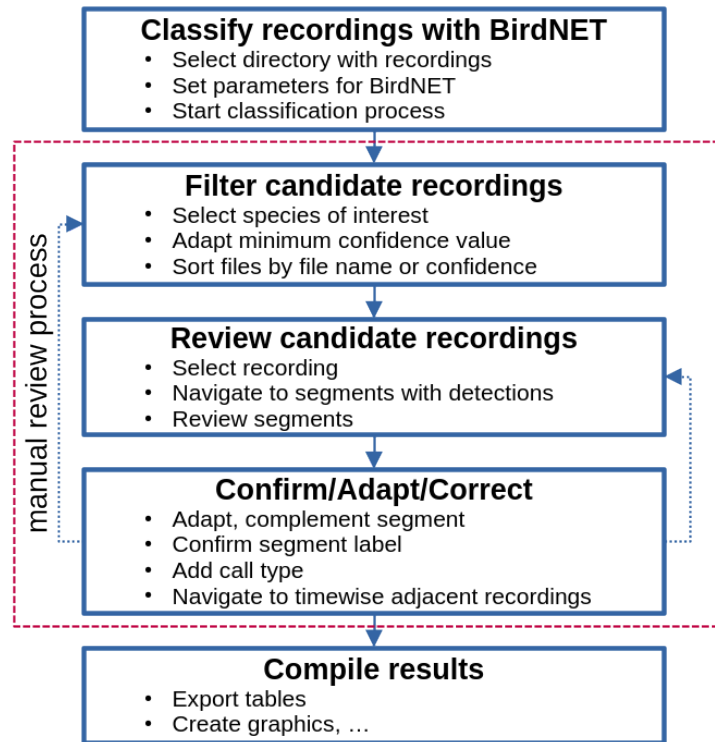
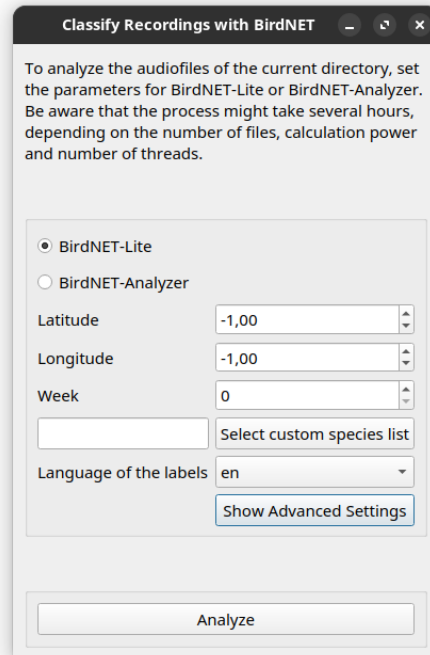


Figure 2: Overview on the classification and review process using the BirdNET classifiers.



In this window the parameters for BirdNET-Lite or BirdNET-Analyzer can be set. The button *Show advanced settings* gives even more options. The meaning of the different parameters is explained in the cheat sheet that you can find over *Help → Cheat Sheet*.

If you are using BirdNET for the first time it is recommended to just set the geographical location with specifying *Latitude* and *Longitude* and provide the time of the year with the week parameter. The other parameters can stay with default values.

After the parameters are chosen, the analysis can be started by clicking on the *Analyze* button. That will classify all the recordings in the current directory including subdirectories. With that it is possible to, for example analyse all the recordings of different locations within one recording session. If you specifically want to use this possibility you can change to an upper directory via the `'../'` in the file list.

Depending on the number of recordings to classify, the CPU power and the number of threads the classification process might take minutes or several hours. Keep that in mind while choosing your parameters. It might also be helpful to test your desired settings on a small test dataset to see if proper results are obtained.

8 Review Classifier Detections

8.1 The output of BirdNET

Before the actual workflow is described the output of BirdNET will be introduced.

Internally BirdNET cuts your recordings into chunks of 3 seconds that overlap more or less depending on the set overlap parameter. For each of these chunks BirdNET produces a list of all classes BirdNET can distinguish (mainly bird species, hereafter species). To each of these species a confidence score ranging from 0 to 1 is attached. With setting the different parameters of BirdNET (for example: minimum confidence, latitude, longitude, week, custom species list) this list is reduced to a minimum of at most a few species.

For interpreting the results it is important to understand that the confidence scores are not calibrated. That leads to two important things:

- The confidence score is not a true probability or certainty value. A value of 0.8 does not mean, that on average the species can be found in 8 out of 10 segments labelled with this certainty value.
- The confidence scores can not be compared across different species. A value of 0.8 can have a completely different meaning for a pygmy owl than for a capercaillie.

8.2 Sorting the List of Files

By default the list of files is sorted by their file names. Usually the recordings include a timestamp in the format Year-Month-Date-Hour-Minute-Second - or similar. Sorting the files by file name means sorting by time in these cases as well. In this mode moving to previous/following recordings in the file list also means moving to timewise adjacent recordings.

You can change the sorting behaviour of the list of files by using the option *Sort files by maximum confidence* in the menu under *View*. This sorts the list of files descending by the highest confidence scores detected in the recording for the selected species. Moving to timewise adjacent recordings does not work in this mode anymore.

8.3 Filtering Audio Files

After the classification process finished you will see that the file icons changed. As described in the section [Coulour Codes](#) the color indicates different states of the segments in the file. Moreover, the grey frame within the icon marks the maximum confidence score detected in the file.

The file icon always refers to the currently selected species. This can be selected after the classification process from the dropdown menu on top of the file list. The values after the species in this list indicate the maximum confidence score found for that species in the audio files of the current directory. If you also tick the box next to that dropdown menu, the file list will reduce to the files in which the selected species are found.

You can further reduce the list of files with changing the value of the slider below the dropdown menu. With that it is possible to exclude all the audio files with a maximum confidence value for the selected species below the selected value.

Use these options to systematically search for audio files with high confidence scores for the specified species (considered “candidate recordings”).

8.4 Manual Review

If a first audio file from the file list is selected you can jump to the first segment labeled with the selected species either with pressing the button *Jump to next mark* below the overview spectrogram or by using the shortcut **Ctrl + Right**.

Use the play buttons to start the playback of the file. Alternatively you can use the shortcut **Space** or **Ctrl + Space** to playback the visible area in the spectrogram or just the selected segment respectively.

The goal of this step is to review all the segments within the file for the selected species.

8.5 Confirm/Change/Correct segments

If the segment is checked it should be confirmed, changed or corrected right after it. Only this way you can avoid to check the same segments and audio files again and again.

For that purpose different options exist:

- To confirm the label of a selected segment use the *Confirm labels* button in the *Controls* area or simply hit the **Enter** key.
- Use your mouse to drag single segments to change their position.
- Use your mouse and the diamonds in the corners of a selected segment to resize it.
- Use the *Delete segment* button or the **Delete** key to delete segments
- Use the *Delete all segment* option from the *Actions* menu or the shortcut **Ctrl + D** to delete every segment in the current file. This can be handy if you want to replace a lot of BirdNET detections in a file with just one segment.

Note: *If there are several species attached to a single segment you can only confirm all the labels. In that case confirm all the labels first and deselect the “wrong” species by simply clicking with the right mouse button on the segment and deselecting the species from the opening menu.*

- If you can neither confirm nor refuse the label BirdNET has given to the segment it might be helpful to assign a custom ‘species’ label. Later, this label can be used to filter out recordings with these doubtful segments. The process how you can add custom labels is described in [Create and Label Segments](#).
- Additionally to the species label PAMalyzer provides the options to assign call type labels. You can change the type of the labels that are shown in the segments by pressing the *SP/CT* button in the controls area or with the **Tab** key. With right clicking on a selected segment a menu opens where you can select call types from a list or add a new call type for the specific species by choosing the *Add calltype* option.

After that process you can go on to the next file using the **Down** arrow key.

8.6 Include temporal adjacent recordings

If a species was confirmed in an audio file it might be a good idea to look for further calls of the same species in the files that were recorded right before or after that. Because of the filtering process it might be that these recordings do not appear in the file list anymore. Nevertheless you can reach those recordings by pressing the **Alt** key together with the **Up** or **Down** keys when moving on to the previous or next audio file. For this option you have to sort the files by name, not by confidence score!

This process might be helpful to gain further ecologically interesting information about the species. Especially faint calls are still a challenge for recognizers like BirdNET.

9 Database

9.1 Choose Database

9.2 Update file paths

9.3 Database design overview

10 The BirdNET Workflow

10.1 Summarise results

If every candidate recording is reviewed the annotations can be exported to an Excel workbook. For that you can use the *Export all segments to excel* option from the *Actions* menu.

This option creates several Excel files in the current directory. The *DetectionSummary_Any_sound.xlsx* contains every segment of all the species within the current directory including subdirectories. Furthermore, individual files for every species are created with the pattern *DetectionSummary_SpeciesName.xlsx*.

In these output files you can find a table in which for every segment the file, the start and end time of the segment as well as the minimum and maximum frequency and call type label are given. In the Excel file for 'any sound' a further column with the species label is attached.

You can use these exported data for example for further analysis or simply to create graphics.

11 Outputs

The PAMalyzer program aims to provide detailed and easy-to-review outputs. An Excel file (with the name 'DetectionSummary' followed by the species selected) will be generated with three sheets of outputs in the same directory as the files. If this file already exists, the new results will be appended to the end of each sheet.

The three sheets of the Excel workbook are:

1. start and end times of each birdcall detected

2. presence/absence of the target species (or set of species) in each recording
3. presence/absence of the target species (or set of species) in each time interval that the user specifies (by default 60 seconds)

In addition to the Excel file, for each audio file PAMalyzer generates an annotation file of the automated detections that the user can open and review in either the main interface or using the *Review Batch Results* option.

12 Reference

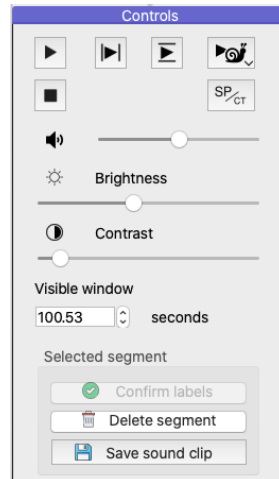
12.1 Interface panels

12.1.1 Overview

12.1.2 Spectrogram

12.1.3 Files

12.1.4 Controls

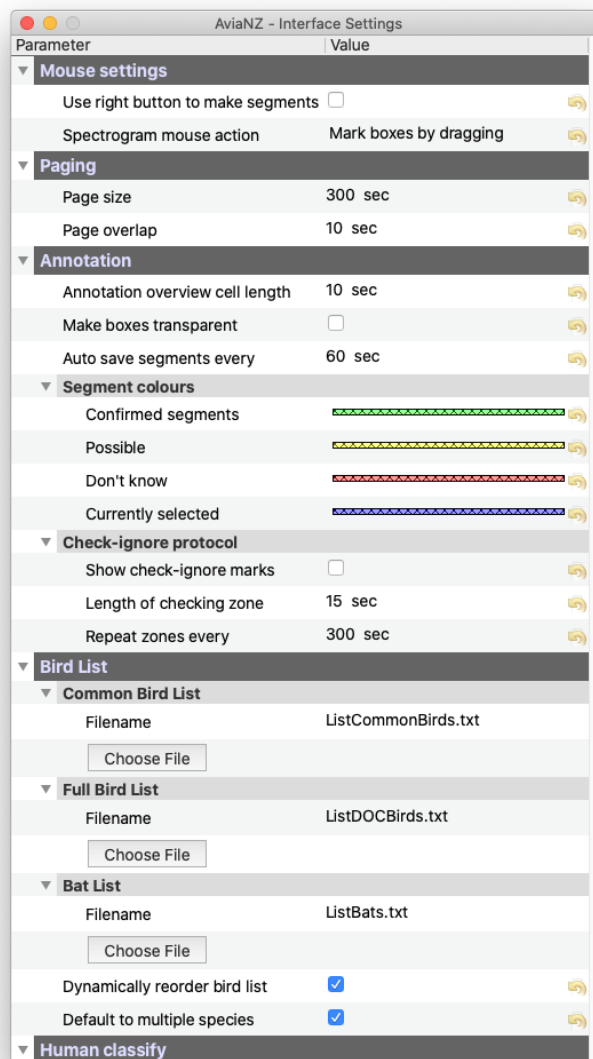


- The buttons at the top of the controls block allow you to play the sounds.
 - The top-left button is a normal *Play* button. The button turns into a *Pause* button while playing. Instead of the buttons you can start and pause the playback using the shortcut **Space**.
 - While the sound is playing, the light blue bar in the spectrogram plot shows where the playback is up to.
 - When the sound is paused, you can drag this bar if you want to hear a particular part of the file. Move the mouse over it, and the bar will go red. Then click and drag (using the mouse button that does not make segments, by default the right button) to move it.

- To stop playback and have the slider return to the start of the visible section, press the *Stop* button.
- When a segment is selected (so that it is blue) you can use the three play buttons to the right of the stop button to play just that small segment.
- The difference between these is:
 - * the one on the left () plays all the frequencies in the audio file (shortcut: **Ctrl + Space**)
 - * the one in the middle () plays only the frequencies highlighted, so that you can isolate particular frequencies of a call. This is particularly helpful when there is high level of background noise in a particular frequency range, such as cicadas, or where there are overlapping calls in different frequency bands.
 - * the one on the right (with the snail: ) lets you play the segment at different speeds. Hold the button down to change it to 2x (twice as fast) or 1/2 or 1/4 speed. Note that the pitch also changes, but it can be interesting or useful to hear all the components of a call.
- You can change the volume of playback using the slider below the buttons.
- The SP/CT button changes whether you control species labels or call type labels of the segments (shortcut: **Tab**).
- The *Brightness* and *Contrast* sliders change the appearance of the spectrograms (2, 7), helping to see more sounds.
- The size of the *Visible window* controls how much of the overview spectrogram (2) appears in the main spectrogram (7).
- The *Confirm labels* button makes a segment label certain. This is useful to confirm labels that were made by an automatic recogniser, or that the user was previously unsure about.
- The *Delete segment* button removes any segment that is selected (blue colour).
- The *Save sound clip* button saves a selected segment as a short audio file.

12.2 Interface settings

- There are quite a few user-selectable options in PAMalyzer, which you can choose using the *Interface settings* menu option in the *View* menu, which will produce the following dialog box:



- These options include:

Mouse settings Swap which mouse button selects segments and which creates them, and change the method of creating segments (described in detail in [Create and Label Segments](#)).

Paging The page is the length of spectrogram loaded into PAMalyzer at one time. By default it is 300 seconds (5 minutes). Note that longer times will require more memory and processing time. The amount of overlap between the pages can also be specified. The aim of the overlap is to make sure that calls aren't missed, and that segments are labelled accurately, at the page limits.

Annotation – The first option in this section changes the length of the boxes in the Overview.

- The next enables you to make the labelling boxes transparent (so that only the outline of the box is shown) if you find it hard to see what is in a segment.
- By default, PAMalyzer save the segments you have made every 60 seconds. This can be changed, for example if you want to do it more often for safety.
- You can also change the colours of segments.
- PAMalyzer has a check-ignore protocol for people who only annotate subsets of recordings. This puts a mark on the spectrogram in places where the user should be annotating the spectrogram, which you can control with these options.

Bird list There are two bird lists in PAMalyzer: the short list of common birds, and a longer list of all possibilities, together with a bat list. You can change these files for others (for example, to include non-New Zealand species).

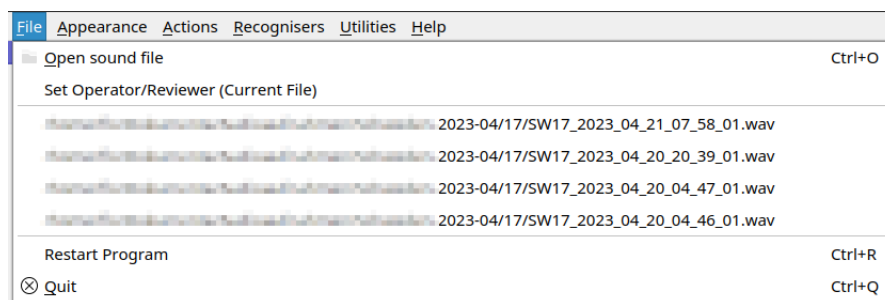
You can also change the default behaviours of dynamically updating the list of common birds, and enable multiple species to be selected for a single segment. This can be useful for labelling things like the dawn chorus, but these segments will not provide good training data for new recognisers.

User Enables you to set the operator and reviewer; this facility can also be found in the *File* menu. You can also change whether or not the PAMalyzer window starts at full screen size, and whether or not the noise data must be completed for all files.

12.3 Menu options

Most of the options for PAMalyzer are found through the menus. There are keyboard shortcuts for many of the menu items, which can be seen in the menus themselves. You can find a reference of shortcuts in the section [Keyboard shortcuts](#)

12.3.1 File Menu



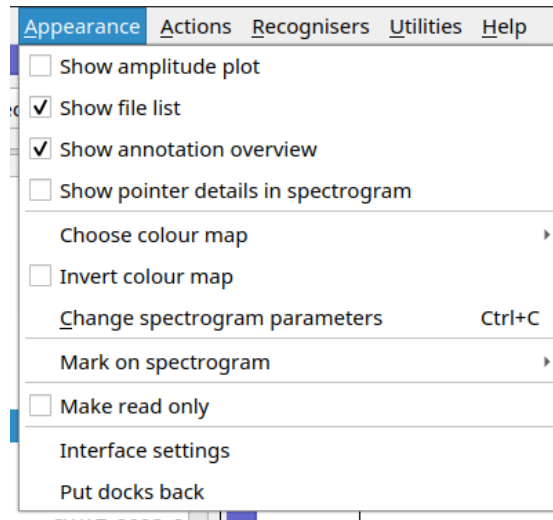
Open sound file Produces a file dialog so that you can choose a new audio file to open.

Set operator (Current File) Enables you to change the operator or reviewer that were specified when you started PAMalyzer. The change only applies to this file.

List of recently used files Makes it easy to revisit recent files.

Quit Does what it says.

12.3.2 View Menu



Changing appearance The first four options in the menu hide or reveal the amplitude plot, file list, annotation overview, and information about where the mouse is pointing in the spectrogram. The ones that are selected are marked with a tick.

Choose colour map This enables the user to select a colour map for the spectrograms they prefer to the default grey one. To adjust the settings for the custom colour map, use the option *Interface settings* on the bottom of the menu.

Invert colour map By default, areas of high energy (frequencies where there is a call or other sound) are shown as the lightest colour, and low energy as dark. This can be swapped over with this option. Note that you will need to change the brightness and contrast using the [Controls](#) area after inverting the colour map.

Change spectrogram parameters A set of options to modify the spectrogram. See the section [Changing the Spectrogram Computation](#) for more details.

Mark on spectrogram There are some features that will help you spot bird calls or recognise them. We currently offer two options here: the fundamental frequency and points of high energy. To select one, choose it from the menu; select it again to remove it. They can help to find particular types of call.

Make read only When reviewing a segmentation it is possible to click on the plots by mistake, adding further segments. This option avoids this problem. It can be turned off by selecting it again. When read only mode is on, the message section at the bottom of the screen says so.

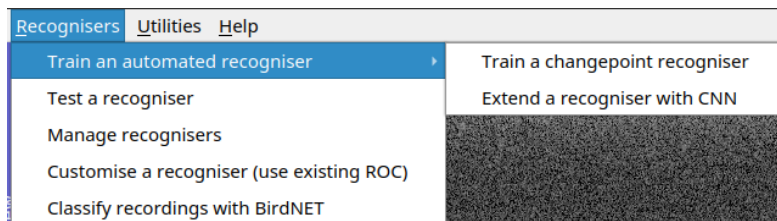
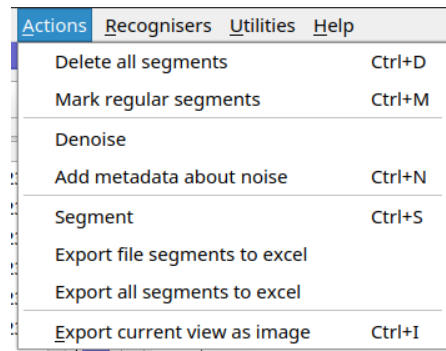
Sort files by maximum confidence This option sorts the list of files by the maximum confidence value of detections for the species selected in the *Files* section. To enable a rank-based review process it excludes confirmed segments from sorting.

Export view Either export the overview spectrogram (from the overview panel) or the detailed spectrogram (from the spectrogram panel, including annotations) as image.

Restore default layout Returns the various screen components to their original layout if they have been moved around.

Interface settings This option produces a dialog that enables the user to customise several things about PAMalyzer. See the section [Interface settings](#) for more details.

12.3.3 Analysis Menu



These options will mostly provide dialog boxes that ask you to make choices.

Classify recordings with BirdNET Choosing this option a window will open where you can set the parameters for BirdNET to classify all your recordings in the current directory and in all subdirectories. It is described in more detail in the section [Classify Recordings with BirdNET](#)

Denoise Runs some programs that try to get rid of the noise in the audio file so that the birdcalls are easier to see. See section [Denoising](#) for more information.

12.3.4 Annotations Menu

Import Annotations

From AviaNZ (.data files) If you have made user annotations in other programs that you want to import, this option will help to do it.

From Raven (selection tables) If you have made user annotations in other programs that you want to import, this option will help to do it.

Export Annotations

To Excel (File) This enables the user to save a summary of the annotation of the currently opened file into an Excel workbook (in the same location as the current audio file), as is described in section [Outputs](#).

To Excel (Directory) This option summarises all the annotations of the current directory (including subdirectories) into Excel workbooks as described in section [Outputs](#).

To database If you want to copy a set of user annotations (for example to have them backed up, or to transfer them to another computer) without the wav files, but with the directory structure preserved, this option will do that.

To .data files

Export Audio

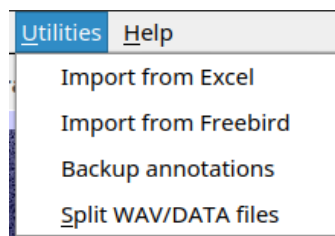
Delete Annotations

File annotations Deletes all annotations from the currently open file

Directory annotations Deletes all annotations in the currently open directory

Mark regular segments Some users find it helpful to have an automatically generated segment box at regular intervals, for example to label all the birds in that segment under some statistical protocol. By default these boxes are made to be 15 seconds long, with one box every 300 seconds (5 minutes) through the file. These options can be changed using the *Check-Ignore protocol* area in the *Interface settings*. There is also an option there to simply mark those areas rather than putting a segment box in place.

Database



Split WAV/DATA files DOC AR4 recorders produce 15 minute audio files, but many other recorders produce far longer recordings. This can mean that you spend a lot of time going forward and backward through spectrogram pages. This option lets you split the audio files into shorter pieces. If you have already processed them, so that they have segments included, there will also be transferred to the new audio files.

12.3.5 Help Menu

Help Gives access to this manual.

Cheat sheet Gives access to a cheat sheet with useful shortcuts and some information about the BirdNET classifier parameters.

About Shows you basic information about PAMalyzer.

12.4 Menu items

12.5 Keyboard shortcuts

Action	Shortcut
General	
Open audio file	Ctrl + O
Quit program	Ctrl + Q
Open the Manual	Ctrl + H
Show About Info	Ctrl + A
Navigation	
Open the previous audio file in the file list	↑
Open the next audio file in the file list	↓
Move backward within the current audio file	←
Move forward within the current audio file	→
Jump to the next page (page size set in Interface settings)	Shift + →
Jump to the previous page	Shift + ←
Jump to the next segment	Ctrl + →
Jump to the previous segment	Ctrl + ←
Navigate to the previous file including hidden files	Alt + ↑
Navigate to the next file including hidden files	Alt + ↓
Playback	
Play / pause the visible spectrogram area	Space
Play only the selected segment	Ctrl + Space
Segment creation	
Apply the label of the previously created segment to the new one (no menu shown)	Shift + drag

Mark the new segment as uncertain (adds a ? to the label, confidence of 50%) **Ctrl + drag**

Editing

Delete the currently selected segment	Delete/Backspace
Delete all segments in the current file	Ctrl + D
Confirm the label of the selected segment	Enter
Toggle between Species and Call-type view (SP/CT button)	Tab

Spectrogram

Change spectrogram parameters	Ctrl + C
Export current view as image	Ctrl + I

12.6 Symbol table

12.7 The Main Interface